

Business Problem Statement

A retail company wants to understand how its customers shop so that it can increase sales, improve customer satisfaction, and build customer loyalty. The company has noticed changes in buying habits based on customer demographics, product categories, and shopping channels (online and offline).

The management team wants to identify the factors that influence customer purchasing decisions and repeat purchases. These factors may include discounts, product reviews, seasonal trends, and preferred payment methods.

The main business question is:

"How can the company use customer shopping data to identify trends, improve customer engagement, and make better marketing and product decisions?"

Deliverables

1. Data Preparation and Modeling (Python)

Clean, preprocess, and transform the raw dataset so it is ready for analysis.

2. Data Analysis (SQL)

Organize the data into structured tables and write SQL queries to generate insights about customer segments, loyalty, and the factors influencing purchases.

3. Visualization and Insights (Power BI)

Create an interactive dashboard that presents key trends and patterns to support data-driven decision-making.

4. Report and Presentation

Prepare a report summarizing the analysis, key findings, and business recommendations. Create a presentation to communicate the insights and suggested actions to stakeholders.

5. GitHub Repository

Maintain a well-organized GitHub repository containing all Python scripts, SQL queries, datasets (if permitted), and Power BI dashboard files.